A REPORT ON

STEER FEEL MODELING OF STEER BY WIRE SYSTEM AND ITS SAFETY FACTORS AND IMPLICATIONS

BY

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Bosch Global Software Technologies,
Coimbatore

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

(April 2024)

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Submitted in partial fulfilment of the
MTech Automotive Electronics degree programme.

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ABSTRACT

X-by-wire technologies is a well innovative research area in the field of automation. This research project is intended to study the modelling and simulation of Steer Feel module of the SbW system and the broad area comprises of the following

- Modelling and simulation: MATLAB to be used to Model and simulate the feedback and control of the Steer Feel
- Risk Assessment: Assessment of risk factor and redundant setup.

X-by Wire is used in Active Safety application, such as Brake-By- Wire and Steer-By Wire and also used in certain applications in the power train such as Throttle-by-Wire. Though the system is much dependable, the haptic feedback from the mechanical counterpart is required to give us a much better user comfort and has a significant safety factor included in it. The dissertation is focused mainly on the Steer Feel Module based modelling and simulation. Scope of this dissertation is to model the Steer Feel function of the SbW and Risk assessment and impact of the Steer Feel on the safety of the whole SbW System.

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ABSTRACT

X-by-wire technologies is a well innovative research area in the field of automation. This research project is intended to study the modelling and simulation of the Steer Feel module of the SbW system and the broad area comprises of the following

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1. INTRODUCTION

The Steering Feedback helps us feel the front wheel forces and the orientation thus making it a device which can both be a sensory device and an actuating device. The feedback is important to judge and whether the stability is fine and whether understeer or oversteer is happening. During the steering feel assessment, the driver decides a quantitative rating about the vehicle dynamics felt in comparison to its main idea of the reference steering system in those conditions. Thus, it is the result of a subjective comparison between the expected perception and the sensed perception. Ideally, steering will have lots of feel and be comfortably weighted. Many over-boosted steering systems hide feedback, making it more inconvenient for a driver to sense how the car is turning or cornering.

Steer-by-wire (SBW) systems rely on electrical or electro-mechanical system for performing its steering function and this system has turned out as a promising next generation automobile steering system. Steer-by-wire replaces the traditional mechanical steering column linkages with steering assist motors, electronic control units and sensors. It further facilitates better steering control as it is dependent on electronic control system. The various benefits associated with automotive steer-by-wire systems include:

- Offers better design flexibility for a vehicle
- Better configuration of a vehicle's steering functionality
- Better fuel efficiency
- A comfortable ride with reduced vibrations•
- Reduced weight due to elimination of mechanical linkages

One of the most eminent factors contributing to steer-by-wire systems market growth is the increasing demand for automation in automobiles along with rising prospects of driverless cars. Moreover, increasing focus on the part of manufacturers to design and develop fuel efficient cars is another important factor complimenting market growth.

Steer feel is very desirable feature for any driver gives more of an organic connection with the road and the vehicle and confidence to the driver that the vehicle is still in control of the driver and doesn't detach the driver from the system of the vehicle.

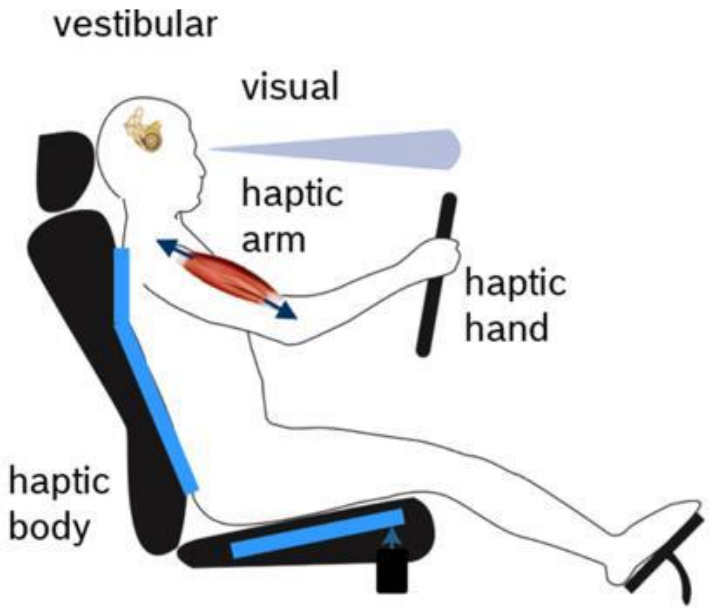


Figure 1: Haptic Feedback Illustration

The only way to created the desired haptic feel is to model using the suspension design and wheelbase and the evaluation is still limited as the real time can be very much needed and vehicle testing feedback has much more precision. Understanding the customer preference is of utmost importance and vehicle design decisions is of utmost importance.

2. MODELLING OF SYSTEMS

The Steering Wheel has haptic feedback as it is connected to the front wheels mechanically and tire reaction Torque is a major Source of the feedback torque to the steering wheel which is reduced considerably in Power Steering – Electric and Hydraulic due to the assistance. The Steering Feel Model is based on the Power Steering Systems and can be derived using the equation.

$$\tau_{feedback} = \tau_{assist} + \tau_{damping} + \tau_{inertia} \quad (1)$$

where $\tau feedback$ – final haptic feedback to the user; $\tau assist$ is the torque t=due to the power assist; $\tau damping$ due to the damping of the steering system and $\tau inertia$ is the torque due to the inertia.

The following torques has been computed individually and the final model is to be derived from each and every Subsection of Torques.

- ## Assist Torque

  The tire Reaction torque felt by the driver is greatly reduced by the Power Steering system and the resultant feedback torque can be said as a function of the weighing function [7]

  W_f and tire reaction torque τtire.

  $$\tau_{assist} = K_{tire} * W_f * \tau_{tire} \qquad (2)$$

  K_{tire} – tire moment gain

  - **Weighing Function** – As the resultant torque of Assist and tire moment depends on the slip angle a function is introduced as a function of Slip angle

    $$Wf = e^{-(\alpha 2/c2)}(1-\gamma)+\gamma \qquad (3)$$

    α - slip angle;

    c – Standard Deviation of Weighing Function; γ- Lower limit of Weighting function.

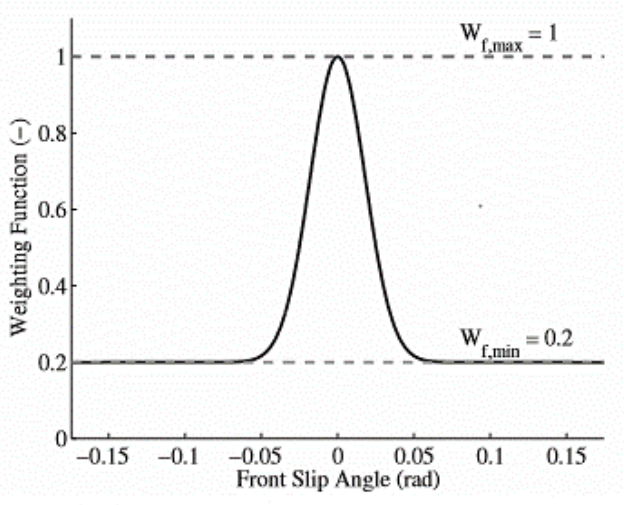


Figure 2: Weighing Function wrt Slip angle

- **Tire Reaction Torque** – Tire reaction torque mainly is based on the three main torques Aligning Torque M_A, Lateral Force ML and Vertical Load M_V.
  - **Aligning Torque M_A :** The lateral force due to the cornering doesn't act at the center of the contact patch but at a distance t_p behind the center and is known as tire pneumatic trial.

    $$M_A = (F_{yl} + F_{yr})\, t_p \qquad (4)$$

    where F_{yl}- Left Front Wheel Lateral Force, F_{yr} – Right Front wheel Lateral Force

    The Pneumatic trial estimation is defined as the decreasing function of the slip angle.

    $$tp = tp0 - (tp0 C\alpha / 3\mu F_Z)^* \, |tan\alpha| \qquad (5)$$

    where tp0 is set to be 1/6 of tire contact patch

    $C\alpha$ – Cornering Fitness

    μ – Cornering Friction

    F_Z – Normal Tire Force (Vertical Force)

  - **Lateral Force ML:**

    $$ML = (Fyl + Fyr)tm \qquad (6)$$

    where tm is the Mechanical trial computation (6) based on the caster angle

    $$tm = r\,tan\mathbf{v} \qquad (7)$$

    v- Caster Angle r -Tire rolling radius

  - **Vertical Load M_V:**

    $$MV = -(Fzl + Fzr)dsin\lambda sin\delta + (Fzl - Fzr)dsinvcos\delta \qquad (8)$$

    Fzr – Vertical Load on Front Right Wheel

    Fzl – Vertical Load on Front Left Wheel

    ν - Caster angle

    λ - Kingpin angle

δ - Steer angle.

d – Lateral offset at the ground

The overall Torque tire reaction from equation (4),(5) and (8)

$$\tau_{tire} = (MA+ML)\ cos\sqrt{\lambda 2+\nu 2+MV} \quad (9)$$

- **Damping Torque**: The Mechanical Conventional System has inherently more intense damping and so it is required to add a damping torque contribution to the Sbw system.The damping can be given by

$$t_{damp} = bs\delta SW \quad (10)$$

bs – Damping of the Steering Column

δSW – Road Wheel Angle Derivative

- **Inertial Torque**: The conventional mechanical system has more inherent inertia and the torque based on inertia can be given by [8]

$$t_{inertia} = Js*\ddot{\delta}S\ W \quad (11)$$

Js – Inertia of the complete steering column

The Road wheel angle derivative is yet to be modelled by precise encoders or a stubbing of the value is yet to be done and tested and will be done on the later changes. The usage of the damping and the inertial torque on the feedback model is yet to be modelled properly due to the significance of the forementioned torque contributions to the steer feel system. The real time damping doesn't have a delay but the inertial torque compensation based on the steer wheel has to be modelled and confirmed for changes.

3. FUNCTIONAL BLOCK DIAGRAM

The Functional level block is attached as follows as per Equation (1)

$$\tau_{feedback} = \tau_{assist} + \tau_{damping} + \tau_{inertia}$$

The following block is an approximation of the simulation based on the mathematical models and is required to be tested and finetuned to be in comparison with the real time steer feel torque values in a conventional mechanical steering system with assist.

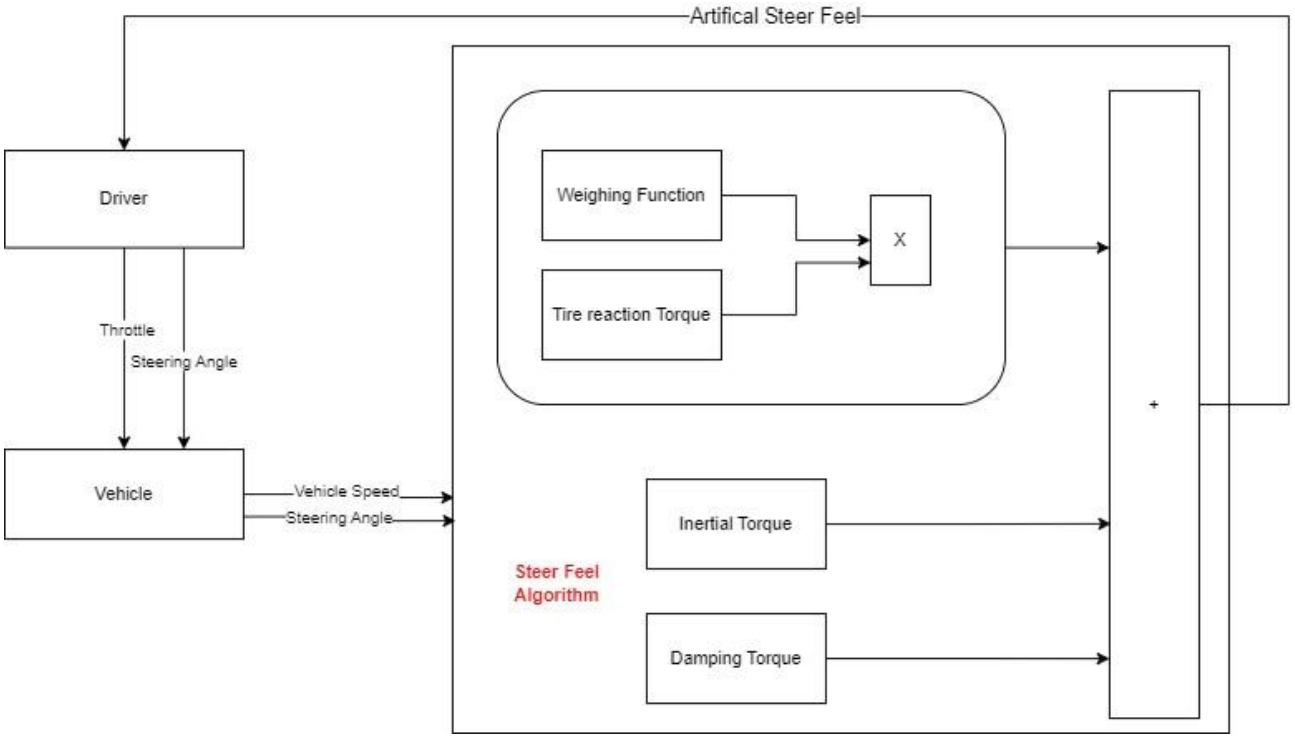


Figure 3: Functional Block Diagram of Steer Feel Model

4. CONSIDERATIONS IN THE MODEL

SNo	Vehicle Parameters	Type of Parameter
1.	Tire lateral forces	Constant
2.	Tire vertical forces	Varying input
3.	Slip angle	Varying input
4.	Tire friction coefficient	Varying input
5.	Tire cornering stiffness	Varying input
6.	Tire rolling radius	Constant
7.	Lateral offset at the ground	Varying input
8.	Kingpin angle	Constant
9.	Caster angle	Constant
10.	Steering angle	Varying input

11.	System damping	Constant
12.	Weighting function	Varying input
13.	Tire moment gain	Varying input
14	Vehicle Speed	Varying input

Table 1: Vehicle Parameters

5. MODELLING OF ALGORITHM

The Modelling of the system needs to be done to validate the algorithm to be developed. For that MATLAB Simulink is used for the Model. The input Parameters mentioned above has been added as a variable and the base workspace has been created and can be changed as a variable.

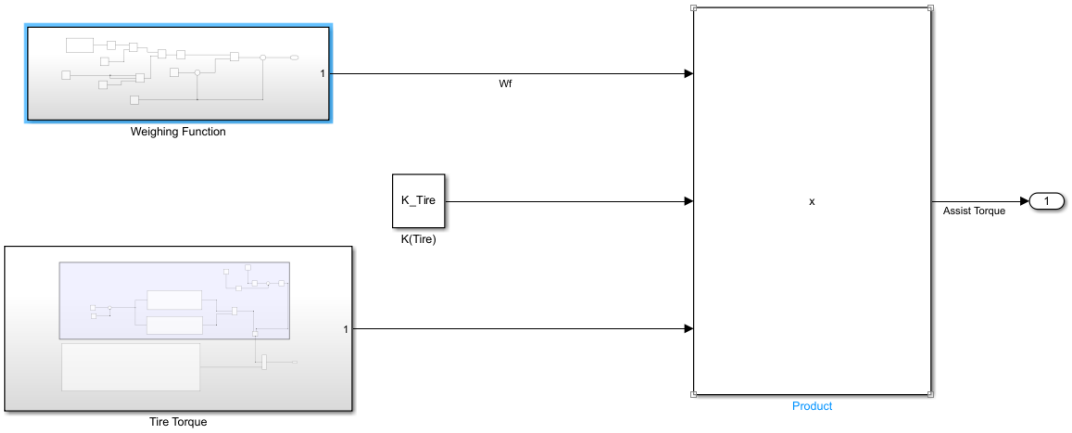


Figure 4: Assist based Torque Model

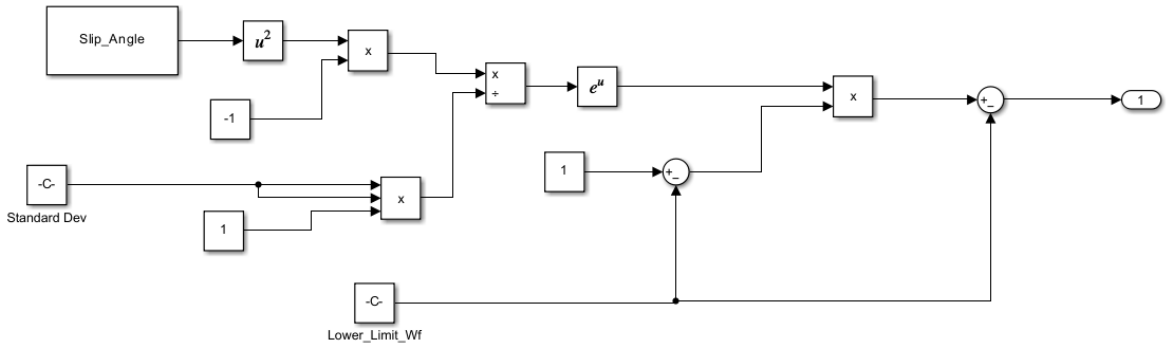


Figure 5: Weighing Function Model

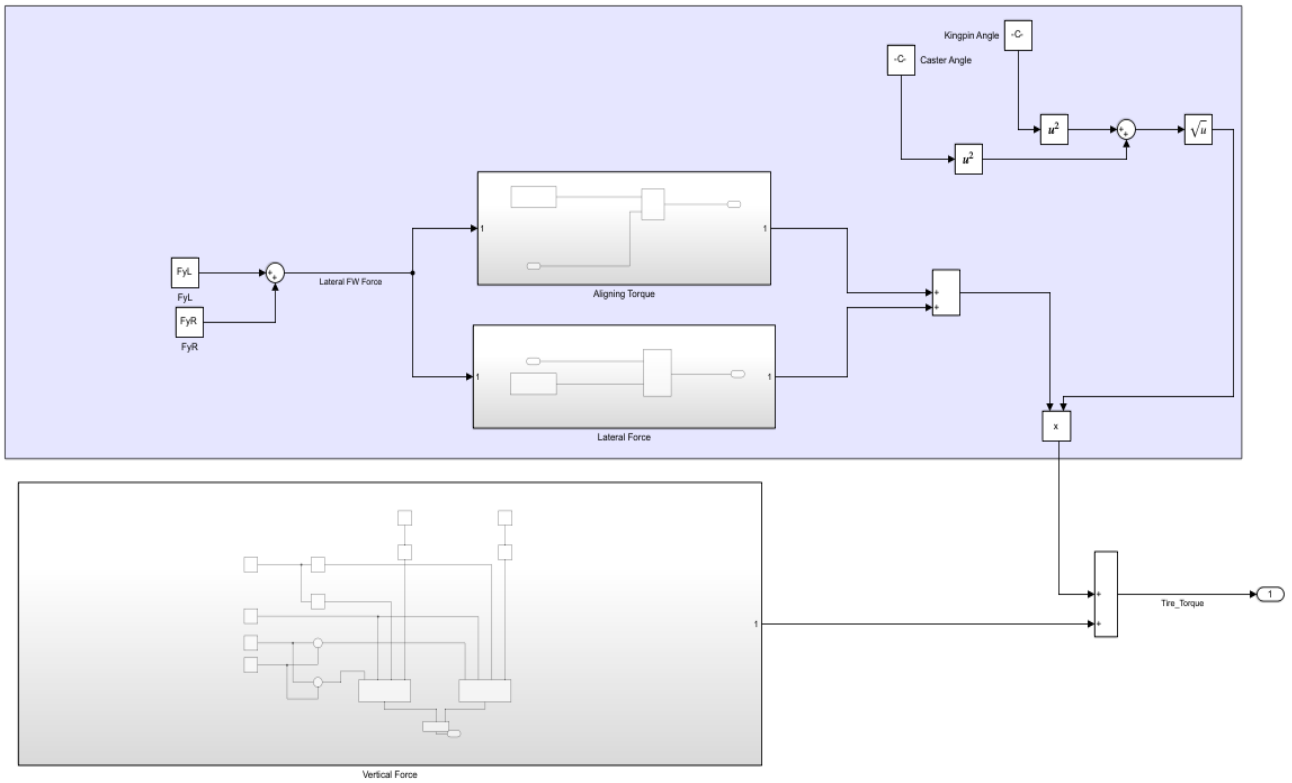


Figure 6: Tire Torque Model

The following model has been tested and the look up tables were selected based on the Vehicle testing characteristics and Lateral Force and Steering Angle were Calculated as a function of the Slip angle. The vehicle speed is kept as a varying input and lookup is between Steering Angle and Slip Angle & Slip Angle and Lateral Force

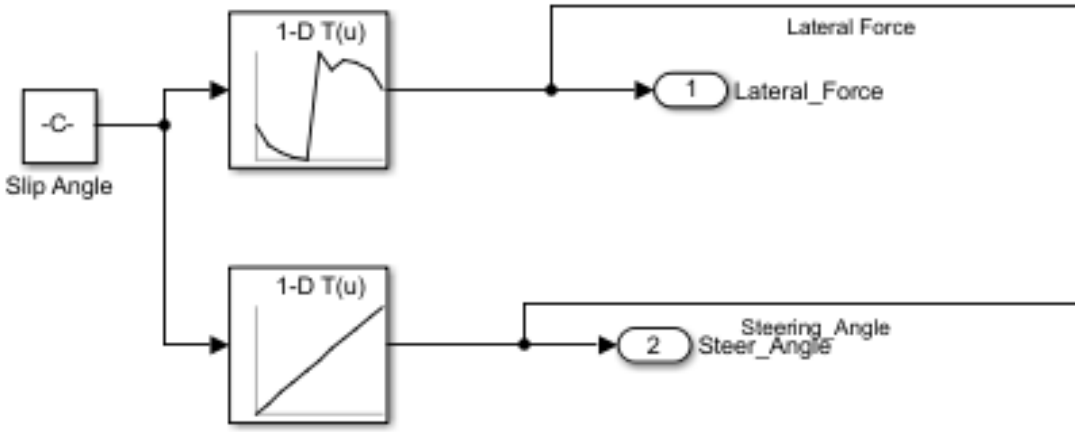


Figure 7: Lookup Table (Internal Source)

The Endstop Torque is also considered and the steering angle is calibrated between 700 and -700 degrees as per the source and a EndStop torque of 3 Nm just before endstop. Endstop is calibrated at 10 degrees before the Mechanical Endstop and the opposing torque will be available and is dependent on the Vehicle Speed. The interpolation data is an approximated version required in the Model Based design and the data cannot be added.

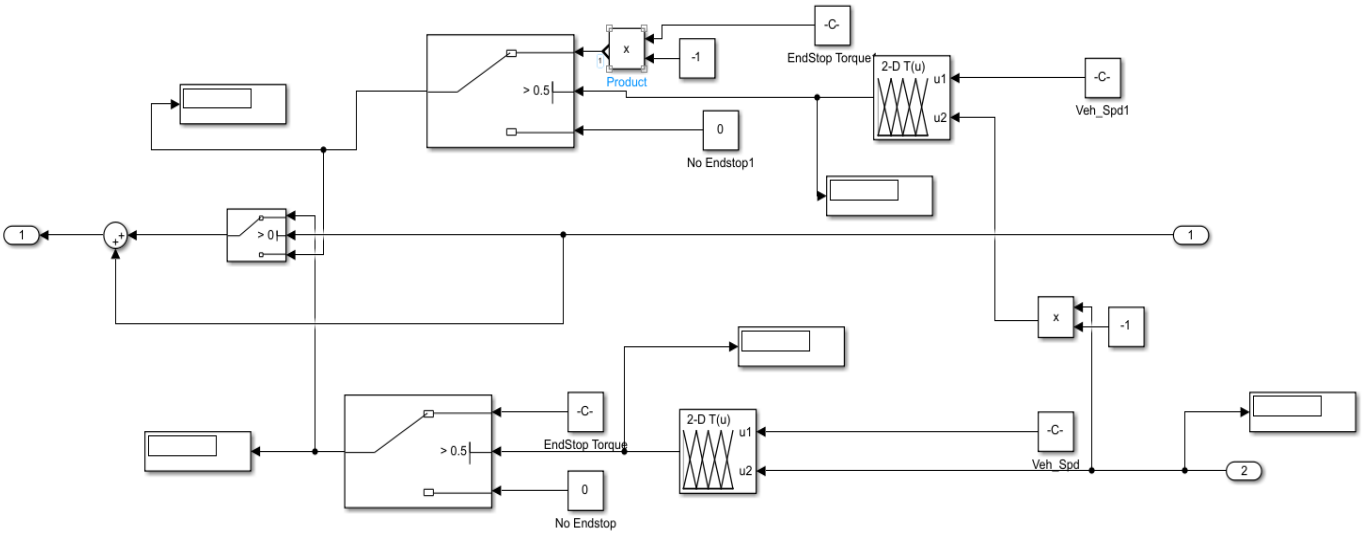


Figure 8: EndStop Algorithm (Internal Source)

6. HARA ANALYSIS

ISO 26262 is an international functional safety standard for the development of electrical and electronic systems in road vehicles. It defines guidelines to minimize the risk of accidents and ensure that automotive components perform their intended functions correctly and at the right time. It also provides an automotive-specific approach for determining risk classes known as ASILs.

HARA analysis (Hazard and Risk Analysis) is a very important assessment in any product according to the Standard ISO 26262. It helps us to determine the ASIL Values considering the following parameters such as severity and the probability of exposure and controllability. It is based on the functional behaviour and thus the detailed hardware and software design need not be known The main objective is

a. to identify and classify the hazardous events caused by malfunctioning behaviour of the item
b. To formulate the safety goals without considering the corresponding ASILS related to the prevention and mitigation of the hazardous events in order to avoid risks.

The Safety Goals has been derived from an internal Source and so the general description of safety goals has been defined in the HARA Analysis attached. The ASIL levels are determined based on the HARA Analysis based on the matrix on severity with probability of exposure and Controllability as follows. The situations are numbered from low to high for all the following parameters.

		Probability class	Controllability class		
			C1	C2	C3
Severity class	S1	E1	QM	QM	QM
		E2	QM	QM	QM
		E3	QM	QM	A
		E4	QM	A	B
	S2	E1	QM	QM	QM
		E2	QM	QM	A
		E3	QM	A	B
		E4	A	B	C
	S3	E1	QM	QM	A
		E2	QM	A	B
		E3	A	B	C
		E4	B	C	D

Figure 9: ASIL Determination Matrix

Hazards		Unintended Steering				
		Controllability		Severity		
Driving Situation	E	Verbal	C	Verbal	S	ASIL
Multi-Lane- entry/exit	E3	Might be tough to control when doing entry exit	C2	AS joining the high speed lane sudden steer can cause issues	S2	B
Multi-Lane -Straight -Line Motion	E4	Control is of less importance in this situation	C1	Straight line motion has less severe effects	S2	A
Multi-Lane -Straight -Line Motion, High Speed	E3	High Speed Control in Straight motion is quite tough to control	C3	High Speed Control in Straight motion is quite tough to control	S3	C
Multi-Lane - Collision Avoidance	E2	Collision unintended steering has less effect as it might lead to lane change at collision avoidance	C2	Collision unintended steering has less effect as it might lead to lane change at collision avoidance	S3	A
Multi-Lane - Lane Changing	E3	Lane Changing control needs precise steering	C2	Lane change might be in the wrong than the intende direction	S3	B
Multi-Lane - Reduced Lane width	E3	Higher risk as the width of the lane is redced	C2	Higher risk as the width of the lane is	S3	B
Multi-Lane - Traction issue	E4	Traction issue changes the Slip and controllability is less	C3	low speeds but at high speeds it is higly severe	S2	C
Multi-Lane - Traffic Congestion	E3	Lesser Controllability in less buffer space	C3	Not lethal while unintened steering as it is at lower speeds	S1	A
Rural Highway - Straight-Line Motion	E4	Control is of less importance in this situation	C1	Straight line motion has less severe effects	S2	B
Rural Highway - Straight-Line Motion, High Speed	E3	High Speed Control in Straight motion is quite tough to control	C3	High Speed Control in Straight motion is quite tough to control	S3	A
Rural Highway - Cornering	E4	Collision unintended steering has less effect as it might lead to lane change at collision avoidance	C2	Collision unintended steering has less effect as it might lead to lane change at collision avoidance	S2	C
Rural Highway - Lane Changing	E4	Lane Changing control needs precise steering	C2	Lane change might be in the wrong than the intende direction	S2	A
Rural Highway -Collision Advance manoeuvre	E2	as it might lead to lane change at collision avoidance	C2	Higher risk as the width of the lane is redced	S3	B
Rural Highway - Traction issue	E4	Traction issue changes the Slip and controllability is less	C3	low speeds but at high speeds it is higly severe	S2	B
Rural Highway - Traffic Congestion	E3	Lesser Controllability in less buffer space	C3	At Traffic Start issue happens at sudden	S2	C
City Traffic - Straight Line motion	E4	Control is of less importance in this situation	C1	Not lethal while unintened steering as it is at lower speeds	S2	A
City Traffic - Cornering	E4	Cornering can have different slip angles at Different Steer angles	C3	Banking can lead to crush at certain	S3	D
City Traffic- Lane Changing	E4	Lane Changing control needs precise steering	C3	Lane Change is risker for pedestrians	S2	D
City Traffic - Collision Advance manoeuvre	E3	as it might lead to lane change at collision avoidance	C3	Collision advance manoeuvre in city traffic is riskier	S3	C
City Traffic -Traffic Congestion	E4	Lesser Controllability in less buffer space	C3	Not lethal while unintened steering as it is at lower speeds	S2	C
City Traffic- Reversing	E3	Reversing Conrollability is affected at unintended Steering	C3	Reversing in Parking is not sever as less vehicular interaction	S1	B
City Traffic - Traction issue	E4	Traction issue changes the Slip and controllability is less	C3	low speeds but at high speeds it is higly severe	S2	C
				Max ASIL	D	
				Safety Goals Met	SG01,SG04,SG05	

Figure 10: HARA Analysis Hazard 1,4,5

Hazards		Unintended Steering				
		Controllability		Severity		
Driving Situation	E	Verbal	C	Verbal	S	ASIL
Multi-Lane- entry/exit	E3	Zero Controllability in all situations	C3	AS joining the high speed lane understeer can cause issues	S3	D
Multi-Lane -Straight -Line Motion	E4	Zero Controllability in all situations	C3	Straight line motion has less severe effects	S2	C
Multi-Lane -Straight -Line Motion, High Speed	E3	Zero Controllability in all situations	C3	High Speed Control in Straight motion is quite tough to control	S3	C
Multi-Lane - Collision Avoidance	E2	Zero Controllability in all situations	C3	Collision unitended steering has less effect as it might lead to lane change at collision avoidance	S3	B
Multi-Lane - Lane Changing	E3	Zero Controllability in all situations	C3	Lane change might be in the wrong than the intende direction	S3	C
Multi-Lane - Reduced Lane width	E3	Zero Controllability in all situations	C3	Higher risk as the width of the lane is redced	S3	C
Multi-Lane - Traction issue	E4	Zero Controllability in all situations	C3	Traction issue doesn't interfere much at low speeds but at high speeds it is higly severe	S2	C
Multi-Lane - Traffic Congestion	E3	Zero Controllability in all situations	C3	Zero Stering has no effect in Traffic Start	S1	A
Rural Highway - Straight-Line Motion	E4	Zero Controllability in all situations	C3	Straight line motion has less severe effects	S2	C
Rural Highway - Straight-Line Motion, High Speed	E3	Zero Controllability in all situations	C3	High Speed Control in Straight motion is quite tough to control	S3	C
Rural Highway - Cornering	E4	Zero Controllability in all situations	C3	Collision unitended steering has less effect as it might lead to lane change at collision avoidance	S3	D
Rural Highway - Lane Changing	E4	Zero Controllability in all situations	C3	Straight ahead when lane Change occurs	S3	D
Rural Highway -Collision Advance manoeuvre	E2	Zero Controllability in all situations	C3	Higher risk as the width of the lane is redced	S3	B
Rural Highway - Traction issue	E4	Zero Controllability in all situations	C3	Traction issue doesn't interfere much at low speeds but at high speeds it is higly severe	S2	C
Rural Highway - Traffic Congestion	E3	Zero Controllability in all situations	C3	At Traffic Start issue happens at sudden start	S2	C
City Traffic - Straight Line motion	E4	Zero Controllability in all situations	C3	Not lethal while unintened steering as it is at lower speeds	S2	C
City Traffic - Cornering	E4	Zero Controllability in all situations	C3	No Steering loss gives issue and understeer occurs	S3	D
City Traffic- Lane Changing	E4	Zero Controllability in all situations	C3	Lane Change is risker for pedestrians	S2	C
City Traffic - Collision Advance manoeuvre	E3	Zero Controllability in all situations	C3	Collision advance manoeuvre in city traffic is riskier	S3	C
City Traffic -Traffic Congestion	E4	Zero Controllability in all situations	C3	Zero Stering has no effect in Traffic Start	S2	C
City Traffic- Reversing	E3	Zero Controllability in all situations	C3	Reversing in Parking is not sever as less vehicular interaction	S1	A
City Traffic - Traction issue	E4	Zero Controllability in all situations	C3	No Steering and slips might not differ	S2	C
				Max ASIL	D	
				Safety Goals Met	SG02	

Figure 11: HARA Analysis Hazard

Hazards		Loss of Assist				
		Controllability		Severity		
Driving Situation	E	Verbal	C	Verbal	S	ASIL
Multi-Lane- entry/exit	E3	Causes Understeer when the Assist is lost in entry conditions	C2	Severity is less as the entry conditions to main line doesn't affect	S1	D
Multi-Lane -Straight -Line Motion	E4	No controllability issue in Straight line motion	C3	Understeer in High speed but no lane change thius severity is low	S2	B
Multi-Lane -Straight -Line Motion, High Speed	E3	High Speed maneuver is affected at Straight line	C3	Understeer happens but severity is still high	S2	B
Multi-Lane - Collision Avoidance	E2	Controllability is lost and partial Assist would help as sudden panic turns will help in assist	C2	The collision avoidance has high risk not to get the required directional Change	S3	A
Multi-Lane - Lane Changing	E3	Controllability is lost and partial Assist would help as sudden panic turns will help in assist	C2	Required Steering at the right time doesn't happen as assist is lost	S2	A
Multi-Lane - Reduced Lane width	E3	The room for steering angle requirement is less	C2	Required Steering at the right time doesn't happen as assist is lost	S2	A
Multi-Lane - Traction issue	E4	Traction happens at low speed and high assist is required	C3	Severity is very less at Traction issu efrom start	S1	B
Multi-Lane - Traffic Congestion	E3	High Assist required in Start/Stop	C3	At Start/Stop Collision and Severiity is less	S1	A
Rural Highway - Straight-Line Motion	E4	No controllability issue in Straight line motion	C3	Understeer in High speed but no lane change thius severity is low	S2	B
Rural Highway - Straight-Line Motion, High Speed	E3	High Speed maneuver is affected at Straight line	C3	Understeer happens but severity is still high	S2	B
Rural Highway - Cornering	E4	Conntrollability issue at Ubdersteer/Oversteer conditions	C2	cornering as steering is not completely Blocked	S3	B
Rural Highway - Lane Changing	E4	Controllability is lost and partial Assist would help as sudden panic turns will help in assist	C2	Required Steering at the right time doesn't happen as assist is lost	S2	A
Rural Highway -Collision Advance manoeuvre	E2	Controllability is lost and partial Assist would help as sudden panic turns will help in assist	C2	The collision avoidance has high risk not to get the required directional Change	S3	A
Rural Highway - Traction issue	E4	Traction happens at low speed and high assist is required	C3	Severity is very less at Traction issu efrom start	S1	B
Rural Highway - Traffic Congestion	E3	High Assist required in Start/Stop	C3	At Start/Stop Collision and Severiity is less	S1	A
City Traffic - Straight Line motion	E4	No controllability issue in Straight line motion	C3	Understeer in High speed but no lane change thius severity is low	S2	B
City Traffic - Cornering	E4	Conntrollability issue at Ubdersteer/Oversteer conditions	C2	cornering as steering is not completely Blocked	S2	B
City Traffic- Lane Changing	E4	Controllability is lost and partial Assist would help as sudden panic turns will help in assist	C2	Required Steering at the right time doesn't happen as assist is lost	S2	A
City Traffic - Collision Advance manoeuvre	E3	Controllability is lost and partial Assist would help as sudden panic turns will help in assist	C2	The collision avoidance has high risk not to get the required directional Change	S3	A
City Traffic -Traffic Congestion	E4	High Assist required in Start/Stop	C3	At Start/Stop Collision and Severiity is less	S1	A
City Traffic- Reversing	E3	Assist not requiored at Reverse/Parking	C1	No severity due to assist loss	S1	QM
City Traffic - Traction issue	E4	Traction happens at low speed and high assist is required	C3	Severity is very less at Traction issu efrom start	S1	B
				Max ASIL	D	
				Safety Goals Met	SG03	

Figure 11: HARA Analysis Hazard 2

The standards have been derived from the ongoing DIN Standard and so complete data of fault analysis cannot be divulged but certain Safety Goals and Final ASIL is derived from the maximum ASIL in all the possible driving situations considered.

Hazard	Safety Goal	ASIL	Safe State for Steer Feel System
Unintended Steering	SG1: Self Steering Prevention during driving	D	X
No Steering availability	SG2: Loss of Steerability to be prevented during Driving	D	-
Loss of Assist	SG3: Loss of handwheel torque shall be prevented during driving	B	X
Unintended Actuator Function	SG4: Unintended blocking of the steering wheel	D	X
Unintended Actuator Function	SG5: Unintended asynchronism between SbW Actuators prevented during Driving	D	-

Table 2: HARA Final ASIL

7. RESULTS AND CONCLUSIONS

The effect of the steering feel model parameters and varying speed on steering feel can be investigated. This knowledge is crucial for understanding and designing steering feel intuitively. In order to control the performance measures, four critical parameters are chosen as follows:

- Change in damping
- Jacking torque spring constant
- Weighting function lower limit
- Tire moment gain

Parameter	Returnability	On-Center Feel	Linearity	Effective Torque Stiffness
Change in Damping	***Increase***	No change	No change	No change
Jacking torque spring constant	No change	***Increase***	No change	No change
Weighting function lower limit	No change	No change	***Increase***	No change
Tire moment Gain	No change	***Increase***	No change	***Increase***

Table 3: Functional Dependency

From changes it has been observed that the change in damping controls the returnability and weighing function lower limit as a direct relation on the linearity of the steer feel torque and on center feel where the slip is zero.The abovementioned signalling indicates the discrete calculations along with Endstop and the values from the continuous signals of the Calculated Torque. The model has simulated the approximated values of the Calculated Steer Torque being high near the center and follows a parabolic curve and decreases towards the end. The directional characteristics are also part of the evaluation.

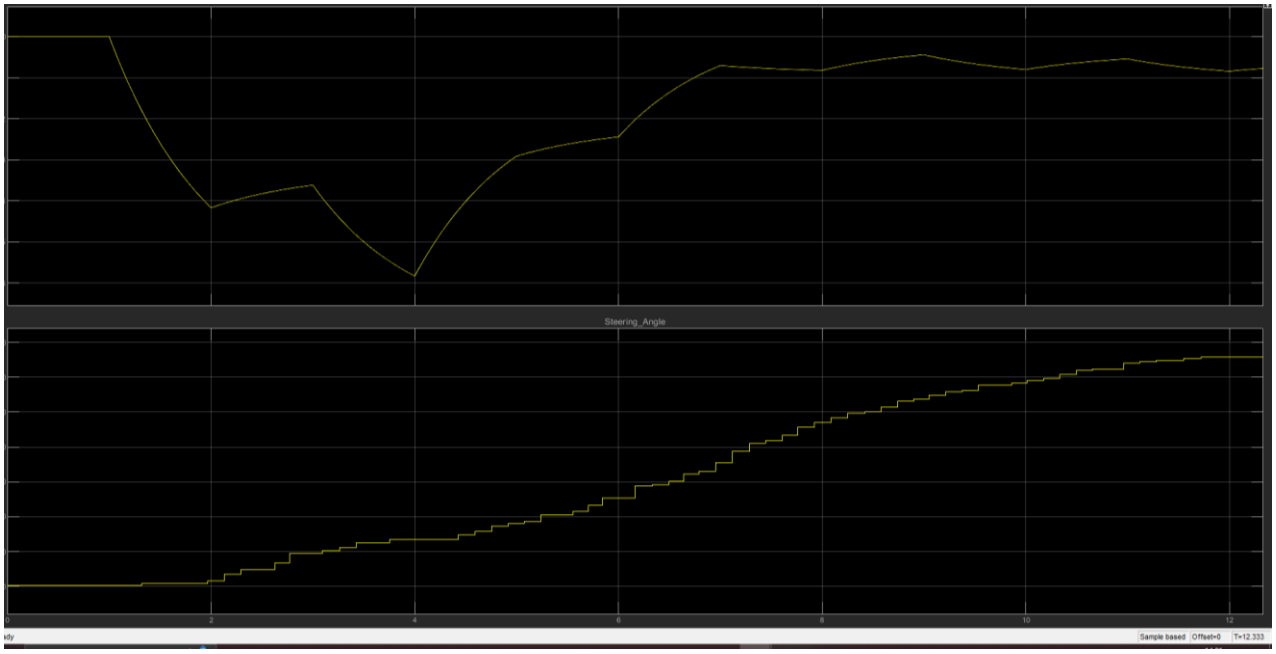


Figure 13: Continuous estimation from Model

The Endstop torque also needs to be added and the continuous addition of the calculation cannot be added as the endstop torque in input as an addition and so the continuous derivation issue is found and the Endstop has been added for the different vehicle speeds and so the endstop start increases at higher vehicle speeds. The discrete addition for the torque contribution has been added as the algorithm for calculation and smoothening cannot be added.

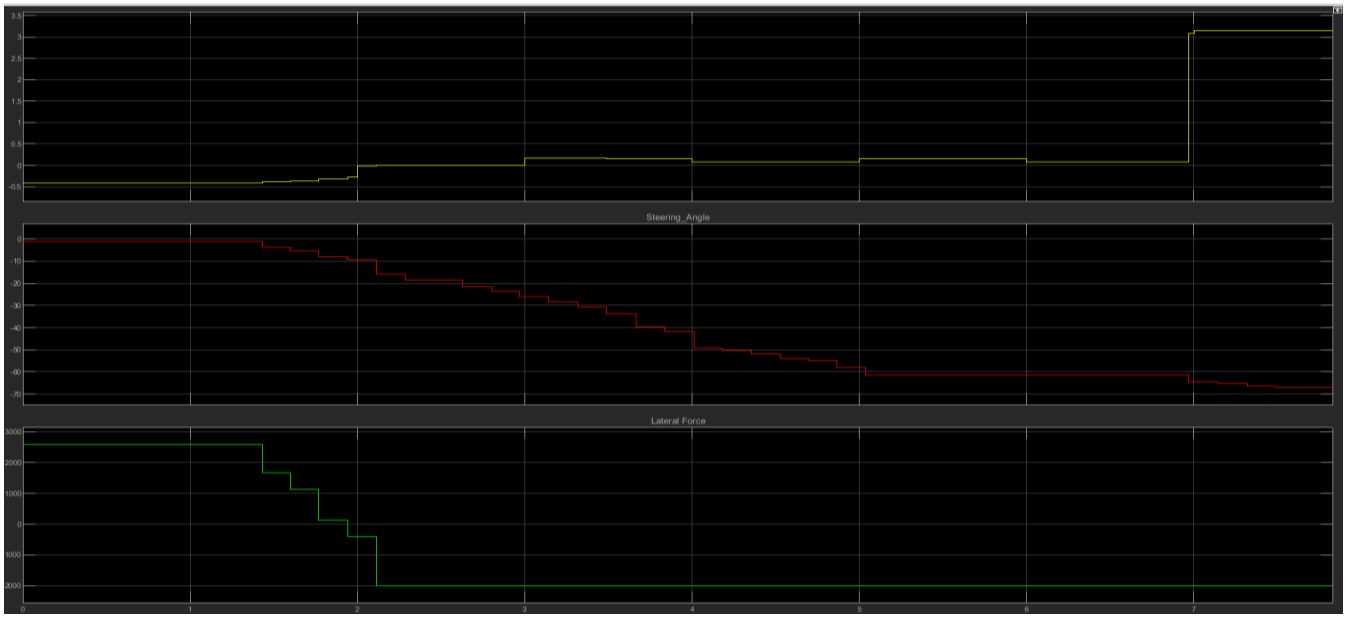


Figure 14: From Steering angle zero to max at Endstop Left Hand Side

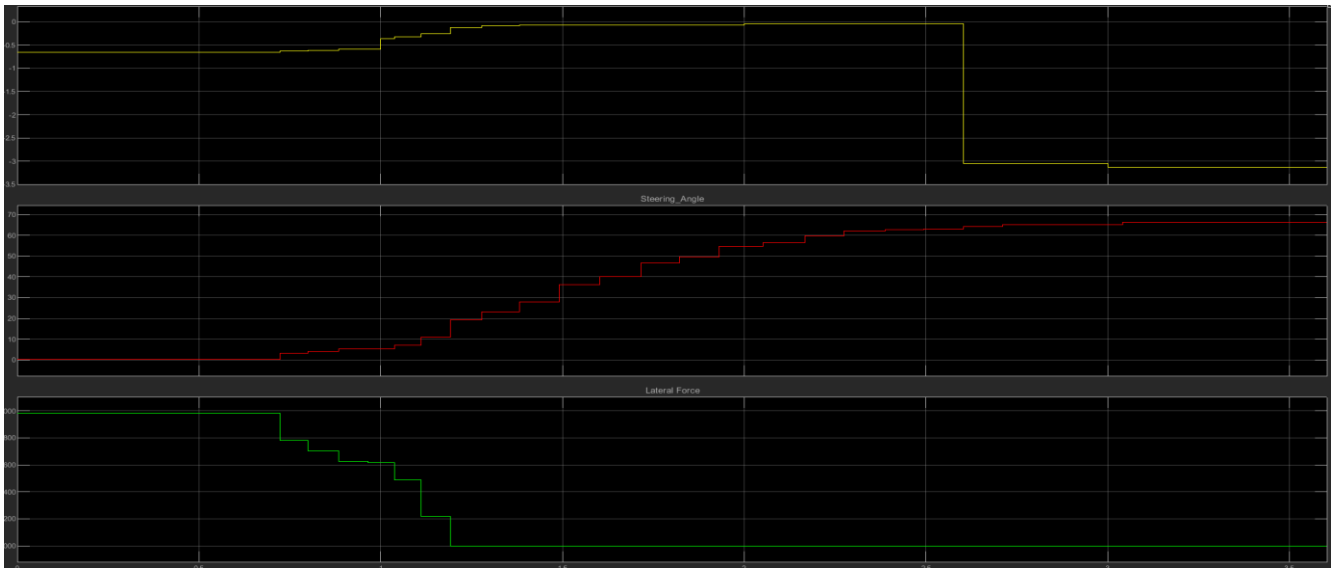


Figure 15: From Steering angle zero to max at Endstop (Right Hand Side)

8. APPENDICES

The following data is about the performance measures mentioned for the steer fee

Returnability: Vehicle Lateral Acceleration at zero hand wheel torque follows an hysteresis curve at a Steering wheel torque and lateral Acceleration

On Center Feel: The Feel while driving high speed in Straight Line driving environments and negotiation of large radius bends at high speeds but low accelerations

Steering Torque Linearity: Linearity represents the level of power assist with a larger linearity indicating a lower power assist effect and a clearer feel of tire dynamics and road properties for the driver.

Torque Stiffness: Effective torque stiffness is the steering torque gradient computed between -20 [%] and +20 [%] of the maximum steering input, defined in the ISO13674-1:2010 standard and, is the stiffness felt when turning.

9. REFERENCES

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10. GLOSSARY

Sbw	Steer By Wire
PS	Power Steering
wrt	With Respect to
HARA	Hazard and Risk Analysis
ASIL	Automotive Safety Integrity Level.
ISO	International Organization for Standardization

Table 4: Glossary

Checklist of Items for the Final Dissertation / Project / Project Work Report

This checklist is to be attached as the last page of the final report.

This checklist is to be duly completed, verified and signed by the student.

1.	**Is the final report neatly formatted with all the elements required for a technical Report?**	**Yes**
2.	Is the Cover page in proper format as given in Annexure A?	**Yes**
3.	Is the Title page (Inner cover page) in proper format?	**Yes**
4.	(a) Is the Certificate from the Supervisor in proper format? (b) Has it been signed by the Supervisor?	**Yes** **Yes**
5.	Is the Abstract included in the report properly written within one page? Have the technical keywords been specified properly?	**Yes** **Yes**
6.	Is the title of your report appropriate? **The title should be adequately descriptive, precise and must reflect scope of the actual work done.** Uncommon abbreviations / Acronyms should not be used in the title	**Yes**
7.	Have you included the List of abbreviations / Acronyms?	**Yes**
8.	Does the Report contain a summary of the literature survey?	Yes
9.	Does the Table of Contents include page numbers? (i). Are the Pages numbered properly? (Ch. 1 should start on Page # 1) (ii). Are the Figures numbered properly? (Figure Numbers and Figure Titles should be at the bottom of the figures) (iii). Are the Tables numbered properly? (Table Numbers and Table Titles should be at the top of the tables) (iv). Are the Captions for the Figures and Tables proper? (v). Are the Appendices numbered properly? Are their titles appropriate	**Yes** **Yes** **Yes** **Yes** **Yes** **Yes**
10.	Is the conclusion of the Report based on discussion of the work?	**Yes**
11.	Are References or Bibliography given at the end of the Report? Have the References been cited properly inside the text of the Report? Are all the references cited in the body of the report	**Yes** **Yes** **Yes**
12.	Is the report format and content according to the guidelines? The report should not be a mere printout of a PowerPoint Presentation, or a user manual. Source code of software need not be included in the report.	**Yes**

Declaration by Student:
I certify that I have properly verified all the items in this checklist and ensure that the report is in proper format as specified in the course handout.

R. Aditya

Place: COIMBATORE
Date: 29/04/2024

Signature of the Student
Name: ADITYA GOPAL R
ID No.: 2022HT65151